

Predicting geometric exposure of particle agglomerates: learned graph descriptors versus conventional structural parameters

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Abstract

Particle agglomerates with different internal arrangements can have different degrees of surface shielding. This study asks whether a small set of structural parameters learned from the particle arrangement can predict mean geometric exposure more accurately than six conventional descriptors, including when those descriptors are used by a nonlinear artificial neural network (ANN). The study comprises 180 agglomerates generated by three growth algorithms, with 50, 100, 200 or 500 particles. Permanent bonds connect the initial particle contacts in a discrete-element method (DEM) calculation. Geometric exposure is the fraction of outward surface-normal rays that are not blocked by another particle, averaged over the agglomerate. A graph neural network (GNN) will learn a short vector of artificial structural parameters from the particles and their connections. Its predictions will be compared with linear regression, a random forest and an ANN using the same six conventional descriptors and the same withheld agglomerates. All 180 local DEM calculations met the stated numerical checks. *[Add the final prediction errors, uncertainty across training runs, selected number of learned parameters, and conclusion after the new ML analysis.]* Geometric exposure measures line of sight; its relationship to diffusion and uptake is outside the present validation.

Keywords: agglomerates; geometric exposure; discrete-element method; structural descriptors; graph neural networks

1 Introduction

Agglomerates consist of primary particles arranged in a larger structure. Particle number and overall size provide useful summaries, but they do not fully describe where particles are located or how they shield one another. A compact description of these differences could help connect detailed particle models with simpler engineering models.

Conventional descriptors are calculated using prescribed formulas, for example the radius of gyration or the number of contacts. An alternative is to learn structural parameters from examples. A graph neural network uses a representation in which particles are points and their connections are links. It can combine information from neighbouring particles to construct a small set of numbers that is useful for predicting a chosen property.

The central research question is: *Can a small set of learned graph-based structural parameters predict mean geometric exposure more accurately than conventional descriptors, even when those conventional descriptors are combined by a nonlinear ANN?* A second question is how many learned parameters are needed and how they relate to familiar measures of size and connectivity. The ANN comparison helps distinguish the benefit of using detailed particle information from the benefit of simply using a more flexible prediction model.

The present study addresses these questions using a controlled synthetic ensemble and a purely geometric target. It does not calculate oxygen concentrations, diffusion rates or reaction rates. *[Introduction to be extended: literature on agglomerate morphology, surface shielding, conventional descriptors and graph-based prediction; establish the specific gap and add verified references.]*

2 Materials and methods

2.1 Study design and agglomerate generation

The ensemble contains $3 \times 4 \times 15 = 180$ independent geometries. The three growth methods are ballistic particle-cluster aggregation (BPCA), balanced ballistic cluster-cluster aggregation (BCCA), and local-filling growth. The particle counts are $N = 50, 100, 200$ and 500 , with 15 realizations for each method and size. Every particle has diameter $d = 1 \mu\text{m}$.

BPCA adds particles to an existing cluster at first contact. BCCA brings clusters together at first contact. Local-filling growth uses a local neighbourhood measure to guide particle placement; detector radii of 3, 6 and 12 particle radii are used, with five independent seeds per setting. Each algorithm defines a general growth mechanism and therefore helps control the range of structures produced. It does not uniquely prescribe the compactness or exposure of each realization; these properties must be measured from the generated coordinates. *[Extend the algorithm descriptions with pseudocode and verified source references.]*

Within each method and particle count, the 15 geometries are ranked by $N^{1/3}/(R_g/r)$, where $r = d/2$. The lowest, middle and highest five are labelled open, intermediate and compact. These labels are relative to that method and size; they are not common porosity thresholds across the ensemble. Continuous measured properties are used for comparisons across methods. The generation methods create tree-like initial contact networks, with $N - 1$ contacts. This restricts the range of connectivity studied.

2.2 Bonded DEM calculations

The discrete-element method follows the translation and rotation of individual particles under applied forces and torques. The present calculations use LAMMPS, with spherical particles and rotational bonds connecting the initial contacts. The bonds resist extension, shear, bending and twisting. They do not break, and no new permanent bonds are formed. Unbonded neighbours interact through Hertz-Mindlin repulsive contacts; the repulsive contact model is not added again to bonded pairs.

The generated geometry defines the stress-free bond reference. Small reproducible random translational and angular velocities perturb this configuration, with total linear and angular momentum removed. Numerical drag then dissipates motion. This tests the implemented mechanical response near the chosen configuration; it does not show that the structure would arise naturally or survive arbitrary loading. The parameters in Table 1 are illustrative rather than fitted to a material.

Acceptance checks require finite outputs, unchanged particle and bond identities, maximum overlap below 0.1% of the diameter, displacement below 1% of the diameter after removing rigid translation and rotation, and final kinetic energy below 0.1% of its initial value. Kinetic energy includes translation and rotation. These are numerical checks, not experimental material validation. A stationary control and a time-step refinement comparison are available for one 500-particle case from each generation method.

Table 1: Principal DEM settings. Full settings are recorded in each case’s configuration file.

Quantity	Value
Particle density	1050 kg m^{-3}
Normal / shear bond stiffness	$10/4 \text{ N m}^{-1}$
Twist / bend bond stiffness	$10^{-13}/10^{-13} \text{ N m}$
Normal / shear bond damping	$10^{-8}/10^{-8} \text{ kg s}^{-1}$
Twist / bend bond damping	$10^{-22}/10^{-22} \text{ kg m}^2 \text{ s}^{-1}$
Unbonded contact Young’s modulus	10^9 Pa
Poisson ratio; restitution; friction	0.3; 0.5; 0.3
Translational / rotational numerical drag	$10^{-8} \text{ kg s}^{-1} / 10^{-22} \text{ kg m}^2 \text{ s}^{-1}$
Initial velocity / angular velocity scale	$10^{-3} \text{ m s}^{-1} / 1000 \text{ rad s}^{-1}$
Time step; number of steps	$10^{-10} \text{ s}; 20000$
Output times	0, 1, $2 \mu\text{s}$

2.3 Geometric exposure as the prediction target

For each particle, $M = 512$ approximately uniform surface directions are constructed using a Fibonacci distribution. A ray starts just outside the surface and travels along the outward normal. Let $b_{im} = 1$ if another particle blocks direction m from particle i , and zero otherwise. Particle exposure and mean agglomerate exposure are

$$a_i = 1 - \frac{1}{M} \sum_{m=1}^M b_{im}, \quad \bar{a}_{\text{geo}} = \frac{1}{N} \sum_{i=1}^N a_i. \quad (1)$$

Both lie between zero and one. A larger value means that more outward directions are unobstructed. This is a surface-normal visibility measure, rather than an integration over all directions from every surface point. It is not a diffusion calculation. Uniformly scaling all coordinates and radii leaves it unchanged.

Exposure is calculated for all three stored snapshots. The final configuration is the planned primary sample for learning, giving 180 independent samples. The intermediate times are used to inspect changes during DEM, not to increase the apparent sample count. Particle geometry and exposure are stored together for visualization. A 2048-ray comparison on three representative 500-particle cases checks sensitivity to angular resolution.

2.4 Conventional structural descriptors

The comparison follows the six descriptors used in the earlier report (Table 2). Here $\mathbf{x}_c$ is the mean particle centre, and z_i is the number of neighbours connected to particle i . For identical spheres, the size measure includes the particles’ own contribution to their mass distribution:

$$R_g^2 = \frac{1}{N} \sum_i \|\mathbf{x}_i - \mathbf{x}_c\|^2 + \frac{3r^2}{5}. \quad (2)$$

The exact rule for identifying connections in the final geometry must be fixed before training and used consistently for graphs and descriptors. Permanent mechanical bonds and geometrical contacts are not automatically identical. *[Specify and verify the final contact tolerance, treatment of slightly separated bonded pairs, and consistency with the earlier report’s radius-of-gyration convention.]*

2.5 Prediction models and learned structural parameters

Linear regression combines descriptors through a weighted sum. A random forest combines several decision trees and can represent nonlinear relationships. An artificial neural network

Table 2: The same six conventional inputs will be supplied to linear regression, random forest and ANN models.

Descriptor	Plain-language meaning
N	Number of primary particles.
R_g/d	Spread of particle mass around the centre, normalized by particle diameter.
N_c	Number of unique particle connections, counting each pair once.
$\rho_c = 2N_c/[N(N-1)]$	Fraction of all possible particle pairs that are connected.
$\bar{z} = 2N_c/N$	Mean number of connected neighbours per particle.
$z_{\max}$	Largest number of connected neighbours of any particle.

(ANN) passes the six descriptors through successive layers of adjustable calculations to predict exposure. Its weights are learned from examples. Unlike linear regression, it can learn nonlinear combinations of the same conventional information.

The graph neural network (GNN) receives the detailed particle graph instead of the six aggregate descriptors. Particle properties are attached to nodes, and pair properties are attached to links. Successive network layers exchange information along the links, allowing each particle’s representation to include information about its neighbourhood. A final aggregation step combines particle information into one representation of the agglomerate.

The *latent vector*,

$$\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_q), \quad (3)$$

is simply a short list of q learned, artificial structural parameters. These parameters are alternatives to prescribed descriptors such as size and coordination. The network chooses them so that, together, they help predict geometric exposure. They are not guaranteed to describe every aspect of structure, and each number need not correspond to one familiar physical property. With $q = 1$, the structure is summarized by one learned number. A small prediction network uses only this vector to estimate exposure; no separate structural inputs bypass it.

The planned graph model follows the earlier report’s geometry-aware architecture: six graph layers, 64 internal features and a comparison of different vector lengths. The earlier implementation must be recovered or reconstructed and checked before the settings are treated as final. The ANN size and regularization, which limits excessive fitting to the training samples, will be selected using training-side validation. *[Add the verified node and link inputs, graph-layer equations, ANN architecture, parameter counts, optimization settings and candidate vector lengths after implementation.]*

2.6 Training, validation and fair comparison

The model learns from a training set. A validation set guides choices such as network size and when to stop training. A test set evaluates prediction on agglomerates that were not used for either purpose. Scaling of inputs is calculated only from training samples and then applied to the withheld samples. The target exposure is never an input.

In five-fold cross-validation, the data are divided into five parts, called folds. Each part is withheld once while the model is developed using the other four. An *out-of-fold prediction* is therefore a prediction for an agglomerate made by a model that did not use that agglomerate for training or model selection. Collecting these predictions produces one independently evaluated prediction per sample. This is more informative about generalization than showing agreement on samples used for fitting.

All models will use identical outer test folds. Choices of architecture, vector length, scaling and stopping time will be made using further splits inside the training portion, not the outer test predictions. Related versions of an agglomerate, including snapshots or rotated copies, must stay together. Primary folds will hold out generation-setting groups, defined by method,

particle count and the detector setting where applicable. Additional tests will withhold an entire generation method and, separately, train on smaller agglomerates and test on $N = 500$. These tests address different questions and will be reported separately. *[Freeze and publish the exact split manifest, inner validation rules and random seeds before fitting.]*

Training curves will show how error changes as model weights are updated. Several random initializations will quantify how much the neural-network results depend on their starting weights. A final model may subsequently be fitted to all data for use and structural interpretation, but its training agreement will not replace the withheld-data evaluation.

2.7 Error measures and interpretation

For n withheld samples with calculated exposures y_j and predictions $\hat{y}_j$, the reported measures are

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_j (\hat{y}_j - y_j)^2}, \quad (4)$$

$$\text{MAE} = \frac{1}{n} \sum_j |\hat{y}_j - y_j|, \quad (5)$$

$$R^2 = 1 - \frac{\sum_j (y_j - \hat{y}_j)^2}{\sum_j (y_j - \bar{y})^2}. \quad (6)$$

RMSE emphasizes larger errors, whereas MAE is the average absolute error. Lower values are better. $R^2 = 1$ indicates perfect prediction; negative values are possible and indicate performance worse than predicting the mean of the evaluated targets. A separate baseline that always predicts the training-set mean will also be reported. Error is expressed in exposure units: an MAE of 0.02 means an average error of two percentage points in exposure.

The learned parameters will be compared with conventional descriptors using scatter plots and correlations. Their ability to be reconstructed from those descriptors will also be examined. Correlation means that quantities vary together; it does not establish a physical cause. Learned coordinates can change sign or scale between independently trained networks. Interpretation will account for this ambiguity and will be distinguished from independent predictive evaluation.

3 Results

3.1 Generated structures and DEM checks

All 180 locally computed cases met the stated numerical checks, retaining exactly three particle snapshots and three bond snapshots. Across the ensemble, final-to-initial kinetic-energy ratios ranged from 3.89×10^{-6} to 2.15×10^{-5} . The largest displacement after removing rigid motion was $2.95 \times 10^{-5}d$, and the largest particle overlap was $9.30 \times 10^{-8}d$. The maximum absolute relative change in radius of gyration was 3.95×10^{-6} (rounded upwards). Thus the chosen perturbation produced only small changes in these configurations. These numbers refer to the archived local campaign; independent runner validation is reported separately when complete.

3.2 Distribution and numerical resolution of exposure

The final-snapshot results are summarized in Table 3. Each row includes all four particle counts. Differences between rows therefore describe the sampled ensembles, rather than a universal ordering of the generation algorithms. In particular, the name local filling should not be interpreted as guaranteeing lower exposure than every other method.

For one 500-particle case per method, increasing the ray count from 512 to 2048 changed the agglomerate mean by less than 4.50×10^{-4} . Individual particle values changed by up to

Table 3: Mean geometric exposure at the final DEM snapshot, calculated with 512 rays per particle. Each method contributes 60 agglomerates.

Method	Minimum	Maximum	Ensemble mean
BPCA	0.3933	0.6498	0.5259
BCCA	0.5452	0.7287	0.6419
Local filling	0.5359	0.7458	0.6599

about 0.0274. This limited check supports the reported mean values for those cases, but is not a convergence bound for all geometries. *[Add representative images coloured by particle exposure and distributions resolved by method and particle count.]*

3.3 Prediction accuracy and comparison with conventional descriptors

[Training on the 180-case dataset has not yet been completed. Populate Table 4 from the saved out-of-fold predictions; add predicted-versus-calculated plots, per-fold errors and variability across training seeds.]

Table 4: Planned comparison on identical withheld samples. Dashes indicate results not yet available, not zero error.

Model and input	RMSE	MAE	R^2
Training-mean baseline	—	—	—
Linear regression, six descriptors	—	—	—
Random forest, six descriptors	—	—	—
ANN, six descriptors	—	—	—
GNN, particle graph	—	—	—

3.4 Number and meaning of learned parameters

[Add the vector-length comparison, selection stability across folds, training curves, learned-parameter correlations and reconstruction from conventional descriptors. Do not select a vector length using the final test errors.]

3.5 Prediction for new sizes and growth methods

[Report the method-holdout and 500-particle tests separately from the primary five-fold results, including error distributions and any failure cases.]

4 Discussion

The decisive comparison is between the GNN and the ANN using six descriptors. Both can learn nonlinear relationships, but only the GNN receives the detailed graph. Better GNN predictions on the same withheld samples would support the value of that additional structural information for this target. Similar performance would support using the simpler descriptor-based model within the studied range. Neither outcome should be assumed before the comparison is completed.

There are important limits to this interpretation. Several conventional descriptors are mathematically related: $\bar{z} = 2N_c/N$ and $\rho_c = \bar{z}/(N-1)$. With a tree network, $N_c = N-1$, so much of their variation is determined by particle count. The six inputs are retained for continuity with the earlier report, but they are not six independent structural measurements. A

weak baseline on this ensemble alone would not establish that all conventional descriptors are inadequate.

The generated structures span three growth mechanisms but do not cover dense networks with many loops. Permanent bonds preserve the initial connectivity, and the selected perturbation is deliberately small. Material parameters have not been calibrated against experiments. The exposure target itself is a geometric measure and excludes fluid motion, diffusion and reaction. Any extension to uptake requires separate transport calculations or experiments.

The earlier 135-agglomerate study is a distinct dataset. Its reported prediction scores are not results for the present 180-case ensemble and are not entered in Table 4. Likewise, its prescribed fractal-dimension and prefactor baseline cannot be transferred directly to algorithms that do not use those same controls. *[Extend the discussion after training: effect size, uncertainty, model complexity, comparison with literature and practical implications.]*

5 Conclusions

A reproducible ensemble of 180 agglomerates has been generated and assessed with bonded DEM. Particle-level geometric exposure and three snapshots per agglomerate are available for analysis and visualization. The manuscript defines a comparison between graph-based learned parameters and conventional descriptors, including a nonlinear ANN using the same six descriptors as the existing baselines. *[Complete the conclusions after training: answer the research question quantitatively, state the supported number of learned parameters, and delimit the range of applicability.]*

Data and code availability

Generation, DEM and exposure-analysis code, generated coordinates and local numerical summaries are maintained in https://github.com/khalifali/PrelimDFG_GNN. The current source summaries are in `results/study180/`; model splits, training code, prediction files and checkpoints will be added when available. *[Add the immutable release identifier and permanent archive for submission.]*

Author contributions, funding and competing interests

[To be completed and confirmed by the authors.]

References

[To be extended with verified literature and software citations. The earlier report's bibliography is retained separately as a starting point.]